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Physically-Consistent Full Dynamic Parameter Identification Based on Nonlinear Friction Model

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Abstract— **Aiming at the shortcomings of the current dynamic parameter identification methods in fitting nonlinear friction characteristics, the satisfaction of physically-consistent constraints, and the efficiency of dynamic calculation of the minimum inertial set, a physical consistent dynamic full parameter identification method based on the nonlinear friction model is proposed. Firstly, according to the characteristics of each component of the robot's dynamic equation, the joint trajectory that can cancel the non-friction moment term during the forward and reverse rotation of the joint is designed, and the friction moment is separated from the measured moment; Secondly, the nonlinear friction model is used to model the friction torque, and the friction parameters are obtained by solving the nonlinear least square problem; Finally, the regularized weighted least squares problem with physically-consistent constraints is solved by convex optimization method, and the full parameter set of robot dynamics including friction parameters and inertia parameters is obtained. The experimental results show that the RMS error of the proposed dynamic parameter identification method is the smallest compared with the CAD numerical calculation method and the penalty optimization identification method.**

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I. INTRODUCTION

The dynamic parameters of the robot link usually include 10 inertial parameters and joint friction model parameters. Common methods to obtain dynamic parameters include a numerical calculation based on CAD software, disassembly measurement based on real robot parts and regression based on robot dynamic model [1], [2]. Among them, the regression method based on the robot dynamics model can not only identify the robot joint friction but also measure the data quickly and have high identification accuracy, is a common method at present.

The problem of inertial parameter identification is usually described as the least square regression problem. However, the regression matrix must be full rank to obtain the unique solution to the least squares regression problem. The *n*-DOF open-chain robot has $10n$ inertial parameters, called full parameter set (FPS) $\boldsymbol{\phi} \in \mathbb{R}^{10n}$, and the column of corresponding regression matrix $\boldsymbol{Y} \in \mathbb{R}^{n \times 10n}$ is not full of rank. Therefore, Gautier, Khalil and Mayeda [3]–[5] proposed the concept and calculation method of base parameter set (BPS) $\boldsymbol{\phi}_B \in \mathbb{R}^{p<10n}$. By expressing the FPS of the robot as BPS, the least square analytical solution of BPS can be obtained. However, because some parameters still appear in linear combinations, they cannot be identified separately. At the same time, BPS can only use the Euler Lagrange (E-L) equation to model the robot's dynamics. However, for robots with complex structures and a high degree of freedom, the computational efficiency is low, and it is not easy to carry out real-time control.

In addition to inertial parameters, the robot joint friction model and the accuracy of friction parameter identification will also significantly impact the robot's dynamic performance. The friction of robot joints is usually complex and has the characteristics of high nonlinearity, multi-parameters, and coupling with non-friction torque terms. It is difficult to separate the friction torque from the obtained joint torque information for separate identification. Therefore, many studies [6], [7] use the simultaneous identification of linear friction and inertia parameters. However, many problems exist in the simultaneous identification of these two parameters by using linear friction model: 1) the linear friction model cannot reflect the nonlinear characteristics of the joint at low speed and the friction torque is discontinuous when the angular velocity crosses zero; 2) since the inertia and friction parameters are identified at the same time under the same framework, the two are coupled, and the parameters influence each other, so it is difficult to obtain better identification results. So many studies [2], [8] use the nonlinear friction model to identify the parameters separately. The existing methods take the difference between the measured and the calculated torque as the joint friction torque for parameter identification, which is not based on the actual friction torque. Therefore, the precision of the friction parameters obtained by fitting is low, and the friction model used does not reflect the Stribeck friction characteristics of the joint at low speed.

Because of the above problems, this paper proposes a physical consistent dynamic full parameter identification method based on the nonlinear friction model. The main contributions of this paper are as follows: (1) the joint motion trajectory is designed to offset the non-friction moment term through the forward and reverse rotation of the joint, and the joint friction moment is separated separately to achieve more accurate modelling and parameter identification of the joint friction moment; (2) It realizes the identification of friction parameters and inertia parameters separately, avoids the coupling of the two when identifying at the same time, and improves the accuracy of the identification results; (3) Considering the physically-consistent of inertial parameters, the identified parameters have both accuracy and rationality.

II. PRELIMINARIES

A. Dynamic Modeling

From the E-L equation, the inverse dynamics equation of the *n*-dof open-chain robot can be obtained as

$$\boldsymbol{\tau} - \boldsymbol{\tau}_f = \boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{G}(\boldsymbol{q}), \tag{1}$$

where $\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}} \in \mathbb{R}^n$ are respectively, the joint angles, angular velocities and angular accelerations, $\boldsymbol{M}(\boldsymbol{q}) \in \mathbb{R}^{n \times n}$ is the

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inertia matrix, $\boldsymbol{C}(\boldsymbol{q})\in\mathbb{R}^{n\times n}$ is the matrix of Coriolis and centrifugal forces, $\boldsymbol{G}(\boldsymbol{q})\in\mathbb{R}^{n}$ is the gravity vector, and $\boldsymbol{\tau},\boldsymbol{\tau}_f\in\mathbb{R}^{n}$ is the joint driving torque and friction torque, respectively.

Equation (1) can be linearly expressed by the inertial parameter $\boldsymbol{\phi}$ of the robot as

$$\boldsymbol{\tau}-\boldsymbol{\tau}_f=\boldsymbol{Y}(\boldsymbol{q},\dot{\boldsymbol{q}},\ddot{\boldsymbol{q}})\boldsymbol{\phi}, \tag{2}$$

where $\boldsymbol{Y}(\boldsymbol{q},\dot{\boldsymbol{q}},\ddot{\boldsymbol{q}})\in\mathbb{R}^{n\times 10n}$ is the regressor matrix, $\boldsymbol{\phi}=[\boldsymbol{\phi}_1^{\mathrm{T}},\cdots,\boldsymbol{\phi}_n^{\mathrm{T}}]^{\mathrm{T}}$ consists of the inertia parameters of each link of the robot, and the inertia parameter $\boldsymbol{\phi}_k$ of the *k* link is expressed as:

$$\boldsymbol{\phi}_k=[m_k,h_{k,x},h_{k,y},h_{k,z},I_{k,xx},I_{k,yy},I_{k,zz},I_{k,xy},I_{k,yz},I_{k,xz}]^{\mathrm{T}}, \tag{3}$$

m_k is the mass of the link, $\boldsymbol{h}_k=[h_{k,x},h_{k,y},h_{k,z}]^{\mathrm{T}}=m_k\boldsymbol{c}_k$ is the first-order mass moment, where $\boldsymbol{c}_k\in\mathbb{R}^3$ is the description of the center of mass vector of the link in the link reference frame $\{B\}$,

$$\boldsymbol{I}_k^B=\begin{bmatrix}I_{k,xx} & I_{k,xy} & I_{k,xz}\\ I_{k,xy} & I_{k,yy} & I_{k,yz}\\ I_{k,xz} & I_{k,yz} & I_{k,zz}\end{bmatrix} \tag{4}$$

is the inertia tensor with respect to the link reference frame $\{B\}$.

B. Physically-consistent Constraints of Inertial Parameters

In order to ensure that the identified parameters have clear physical meanings, the physically-consistent constraints used in this paper as

$$m_k>0,\ \boldsymbol{I}_k^B\succ 0. \tag{5}$$

It is also necessary to ensure that the principal moment of inertia $(\lambda_{k,1},\lambda_{k,2},\lambda_{k,3})$ of the inertia tensor $\boldsymbol{I}_k^B$ satisfies the triangle inequality:

$$\lambda_i+\lambda_j>\lambda_l,\forall\{i,j,k\}=\{1,2,3\}. \tag{6}$$

To ensure that the global optimal solution of inertial parameters is obtained under the convex optimization framework, constraints (5) and (6) are rearrange as LMI form [7]:

$$\boldsymbol{P}(\boldsymbol{\phi}_k)=\begin{bmatrix}(\mathrm{tr}(\boldsymbol{I}_k^B)\boldsymbol{1}_3/2-\boldsymbol{I}_k^B) & \boldsymbol{h}_k\\ \boldsymbol{h}_k^{\mathrm{T}} & m_k\end{bmatrix}\succ 0. \tag{7}$$

Therefore, the necessary and sufficient physically-consistent condition of $\boldsymbol{\phi}_k$ can be expressed as the positive definiteness of matrix $\boldsymbol{P}(\boldsymbol{\phi}_k)$.

Meanwhile, this paper also takes geometric consistency constraints into consideration for the identified parameters. The corresponding constraints can be expressed as:

$$\mathrm{tr}(\boldsymbol{P}(\boldsymbol{\phi}_k)\boldsymbol{Q}_k)\geq 0, \tag{8}$$

where $\boldsymbol{Q}_k$ is a 4×4 symmetric matrix, which is used to describe the topological structure of the boundary ellipsoid.

III. Dynamic Parameter Identification

A. The Friction Identification Trajectory and Friction Parameter Identification

To identify the friction parameters individually, it is first necessary to separate the friction torque from the obtained joint torque information. According to the robot inverse dynamics Eq. (1), it can be known that: given a set of joint angles $\boldsymbol{q}$, the inertia moment $\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}}$ is only related to its joint angular acceleration $\ddot{\boldsymbol{q}}$; Coriolis moment and centrifugal torque $\boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}}$ are related to its joint angular velocity $\dot{\boldsymbol{q}}$ and satisfy the equation $\boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}}=\boldsymbol{C}(\boldsymbol{q},-\dot{\boldsymbol{q}})(-\dot{\boldsymbol{q}})$; and the gravitational moment $\boldsymbol{G}(\boldsymbol{q})$ is only related to the joint angle $\boldsymbol{q}$. And the robot joint friction usually behaves as an odd function of the joint angular velocity $\dot{\boldsymbol{q}}$. In order to obtain the friction torque at different speeds separately, the joint friction identification trajectory is constructed in this paper, so that the inertia moment, Coriolis moment and centrifugal torque and gravitational torque are cancelled in the forward and reverse motion of the joint, and only the friction torque $\boldsymbol{\tau}_f$ is retained. The friction identification trajectory of the *k* joint is as follows:

$$q_k=\begin{cases}-a_k t^2+q_{k,0}, & 0<t<\mathrm{step}_k\\ a_k(t-2\mathrm{step}_k)^2+b_k, & \mathrm{step}_k\leq t<3\mathrm{step}_k\\ -a_k(t-4\mathrm{step}_k)^2+q_{k,0}, & 3\mathrm{step}_k\leq t\leq 4\mathrm{step}_k\end{cases}, \tag{9}$$

where $\mathrm{step}_k=\lambda_{k,vec}\dot{q}_{k,\max}/(\lambda_{k,acc}\ddot{q}_{k,\max})$ is the time of the acceleration/deceleration stage, $\lambda_{k,vec},\lambda_{k,acc}\in[0,1]$ are the percentage of joint maximum speed and acceleration, $q_{k,0}\in\mathbb{R}$ is the initial joint angle, $\dot{q}_{k,\max},\ddot{q}_{k,\max}\in\mathbb{R}$ are the maximum speed and acceleration of the joint. a_k,b_k are respectively:

$$a_k=-\lambda_{k,vec}\ddot{q}_{k,\max}/2,\ b_k=-2a_k\mathrm{step}_k^2+q_{k,0}. \tag{10}$$

The initial position of the joint can be adjusted by $q_{k,0}$, and the speed range of the joint can be adjusted by $\lambda_{k,vec},\lambda_{k,acc}$.

The friction moment of the robot joint has the characteristic of symmetry about the origin, i.e., $\boldsymbol{\tau}_f(\dot{\boldsymbol{q}})=-\boldsymbol{\tau}_f(-\dot{\boldsymbol{q}})$. When the joints rotate in the opposite direction with the same angular velocity, they will produce friction torques in opposite directions but equal in magnitude. Therefore, the robot moves according to the above joint friction identification trajectory. Using the friction torque and the dynamic characteristics of the robot joint. By subtracting the total joint torque generated by the joint moving in the same position and two opposite speed directions, we can obtain:

$$\begin{aligned}&\boldsymbol{\tau}(\boldsymbol{q}(t))-\boldsymbol{\tau}(\boldsymbol{q}(4\mathrm{step}-t))\\ &=\boldsymbol{M}(\boldsymbol{q}(t))\ddot{\boldsymbol{q}}(t)+\boldsymbol{C}(\boldsymbol{q}(t),\dot{\boldsymbol{q}}(t))\dot{\boldsymbol{q}}(t)+\boldsymbol{G}(\boldsymbol{q}(t))\\ &+\boldsymbol{\tau}_f(\dot{\boldsymbol{q}}(t))-\boldsymbol{M}(\boldsymbol{q}(4\mathrm{step}-t))\ddot{\boldsymbol{q}}(4\mathrm{step}-t)\\ &-\boldsymbol{C}(\boldsymbol{q}(4\mathrm{step}-t),\dot{\boldsymbol{q}}(4\mathrm{step}-t))\dot{\boldsymbol{q}}(4\mathrm{step}-t)\\ &-\boldsymbol{G}(\boldsymbol{q}(4\mathrm{step}-t))-\boldsymbol{\tau}_f(\dot{\boldsymbol{q}}(4\mathrm{step}-t))\\ &=2\boldsymbol{\tau}_f(\dot{\boldsymbol{q}}(t)),\end{aligned} \tag{11}$$

where $\boldsymbol{\tau}_f(\dot{\boldsymbol{q}}(t))\in\mathbb{R}^n$ is the joint friction torque at time *t*, and $\boldsymbol{\tau}(\dot{\boldsymbol{q}}(t))\in\mathbb{R}^n$ is the joint measurement torque collected by the robot running the friction identification trajectory. So that a separate friction torque can be obtained

$$\boldsymbol{\tau}_f(\dot{\boldsymbol{q}}(t))=[\boldsymbol{\tau}(\boldsymbol{q}(t))-\boldsymbol{\tau}(\boldsymbol{q}(4\mathrm{step}-t))]/2. \tag{12}$$

In order to accurately fit the friction torque, the following nonlinear friction model is used to represent the joint friction $\tau_{k,f}$ of the robot:

$$\begin{aligned}\tau_{k,f}(\dot{q}_k)=&\alpha_{k,1}(\tanh(\alpha_{k,2}\dot{q}_k)-\tanh(\alpha_{k,3}\dot{q}_k))\\ &+\alpha_{k,4}\tanh(\alpha_{k,5}\dot{q}_k)+\alpha_{k,6}\dot{q},\end{aligned} \tag{13}$$

where the friction characteristics of the *k* joint are determined by the parameter $\alpha_{k,i}\in\mathbb{R},i=1,2,\cdots,6$.

The identification of friction parameters can be described as the following optimization problem:

$$\min_{\alpha_k}\|\tau_{k,f}-\hat{\tau}_{k,f}\|^2, \tag{14}$$

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Since the friction model (13) is a nonlinear model, the identification of friction parameters is a nonlinear least square fitting problem, which can be solved by the Levenberg-Marquardt method.

B. Excitation Trajectory

In order to excite the inertial parameters of the robot as much as possible, this paper adopts the Fourier series trajectory, which has many advantages such as smooth derivation and no speed mutation to construct the excitation trajectory for the identification of the inertial parameters of the robot. The excitation trajectory of the k joint is as follows:

$$q_k(t)=\sum_{l=1}^{L}\left(\frac{a_{k,l}}{l\omega_f}\sin(l\omega_f t)-\frac{b_{k,l}}{l\omega_f}\cos(l\omega_f t)\right)+q_{k,0}, \quad (15)$$

where L is the harmonic order, ω_f is the angular frequency, and $a_{k,l}, b_{k,l}, q_{k,0}$ are the design parameters related to the trajectory amplitude, phase and offset.

C. Inertial Parameter Identification

Control the robot to move according to the constructed excitation trajectory, collect $M \gg n$ sets of torque, angle, and angular velocity data, and filter the collected data. And use the above sample data for the identification of inertial parameters, for each sample data ($\boldsymbol{\tau}_i, \boldsymbol{q}_i, \dot{\boldsymbol{q}}_i, \ddot{\boldsymbol{q}}_i, i=1,\cdots,M$), there are

$$\boldsymbol{\tau}_i-\hat{\boldsymbol{\tau}}_{i,f}=\boldsymbol{Y}(\boldsymbol{q}_i,\dot{\boldsymbol{q}}_i,\ddot{\boldsymbol{q}}_i)\boldsymbol{\phi}, \quad (16)$$

where $\hat{\boldsymbol{\tau}}_{i,f}$ is the joint friction torque obtained by the above independent identification.

All the data obtained by Eq. (16) can be expressed in the form of the following large overdetermined linear equations:

$$\boldsymbol{b}=\boldsymbol{A}\boldsymbol{\phi}, \quad (17)$$

where

$$\boldsymbol{b}=\begin{bmatrix}\boldsymbol{\tau}_1-\hat{\boldsymbol{\tau}}_{1,f}\\ \vdots \\ \boldsymbol{\tau}_M-\hat{\boldsymbol{\tau}}_{M,f}\end{bmatrix}, \boldsymbol{A}=\begin{bmatrix}\boldsymbol{Y}(\boldsymbol{q}_1,\dot{\boldsymbol{q}}_1,\ddot{\boldsymbol{q}}_1)\\ \vdots \\ \boldsymbol{Y}(\boldsymbol{q}_M,\dot{\boldsymbol{q}}_M,\ddot{\boldsymbol{q}}_M)\end{bmatrix}. \quad (18)$$

The unrealistic inertial parameters may lead to non-convergence of the control based on the robot dynamics model, resulting in erroneous motion trajectories [9]. Therefore, physically consistent constraints (7) and (8) are introduced in the solution process to constrain the inertial parameters within the feasible domain. In summary, this paper adopts a regularized weighted least squares dynamic parameter identification model with physically-consistent constraints (7) and (8), which can be expressed as:

$$\min_{\boldsymbol{\phi}} \|\boldsymbol{A}\boldsymbol{\phi}-\boldsymbol{b}\|^2_{\Sigma^{-1}}+\gamma\|\boldsymbol{\phi}-\boldsymbol{\phi}_0\|^2_2, \quad (19)$$

where regular term $\|\boldsymbol{\phi}-\boldsymbol{\phi}_0\|^2_2$ is used as a penalty to measure the error between the estimated value $\boldsymbol{\phi}$ and the nominal value $\boldsymbol{\phi}_0$. The weighting matrix $\boldsymbol{\Sigma}$ is calculated by non-regular parameter estimation. The choice of the regularization factor γ depends on the uncertainty of the nominal value $\boldsymbol{\phi}_0$ relative to the measured data.

D. Solution of Inertial Parameter Optimization Model

First, the covariance matrix is solved without regularization penalty and weighting matrix. The nominal values of the inertial parameters of the robot and the bounding ellipsoid $\boldsymbol{Q}_k$ of each link are obtained from the CAD model. Set the regularization factor $\gamma=0$ to solve the optimization model:

$$\min_{\boldsymbol{\phi}} \|\boldsymbol{A}\boldsymbol{\phi}-\boldsymbol{b}\|^2_2, \quad (20)$$

get a set of robot inertia parameters $\hat{\boldsymbol{\phi}}$. The calculated torque $\hat{\boldsymbol{\tau}}=\boldsymbol{Y}(\boldsymbol{q},\dot{\boldsymbol{q}},\ddot{\boldsymbol{q}})\hat{\boldsymbol{\phi}}$ is obtained according to the obtained inertial parameters, and the residual between the measured torque $\boldsymbol{\tau}$ and the calculated torque $\hat{\boldsymbol{\tau}}$ is obtained by $\Delta\boldsymbol{\tau}=\boldsymbol{\tau}-\hat{\boldsymbol{\tau}}$. $\Delta\boldsymbol{\tau}$ is a $nM\times1$ dimensional vector, which can be reshaped as a $n\times M$ matrix called reshaped residual $\boldsymbol{E}$ here. The variance of the error vector is estimated as follows:

$$\boldsymbol{\Lambda}_{jj}=\|\boldsymbol{E}_j\|^2_2/(M-10n), \quad (21)$$

where $\boldsymbol{\Lambda}_{jj}$ is the j element of the diagonal matrix $\boldsymbol{\Lambda}$ and $\boldsymbol{E}_j$ is the j row of the matrix $\boldsymbol{E}$. This gives the covariance matrix:

$$\boldsymbol{\Sigma}=M\,\mathrm{diag}(\Lambda\ \cdots\ \Lambda)\in\mathbb{R}^{nM\times nM}. \quad (22)$$

Then, set the regularization factor γ according to the uncertainty of the nominal value $\boldsymbol{\phi}_0$ relative to the inertial parameters of the real robot and the noise level of the measured data, take the covariance matrix $\boldsymbol{\Sigma}$ as the weighting matrix, solve the optimization conditions (19), (20), and obtain the final inertial parameter value $\dot{\boldsymbol{\phi}}_*$.

IV. Experiment 1: Identification and Verification of Friction Parameters

The experimental equipment used in this paper is the Franka Emika Panda robot (Panda robot) and its MDH parameters are shown in Fig 1. The Panda robot has $n=7$ revolute joints, and each one is equipped with torque sensors.



Joint	a_k	α_k	d_k	θ_k
1	0	0	0.333	θ_1
2	0	$-\pi/2$	0	θ_2
3	0	$\pi/2$	0.316	θ_3
4	0.0825	$\pi/2$	0	θ_4
5	-0.0825	$-\pi/2$	0.384	θ_5
6	0	$\pi/2$	0	θ_6
7	0.088	$\pi/2$	0	θ_7

Figure 1 Franka Emika Panda robot and MDH parameters

A. Identification of Friction Parameters

According to Eq. (9), the joint trajectory is designed to offset the non-friction moment term through forward and reverse rotation. At the same time, under the condition that the joint restrictions are met, the parameters of the k joint are adjusted (as shown in Tab. 1), so that the identification trajectory can cover the entire joint speed range as much as possible.

TABLE I Friction Identification Trajectory Parameters

Joint k	**trajectory 1, 2**		**trajectory 1**	**trajectory 2**
	$\lambda_{k,vec}$	$\lambda_{k,acc}$	$q_{k,0}$	
1	0.5	0.03	2.8973	-0.1027
2	0.5	0.05	1.7628	1.2628
3	0.5	0.03	2.8973	0.8973
4	0.5	0.04	-0.0698	-0.6698
5	0.4	0.03	2.8973	0.8973
6	0.4	0.03	3.7525	2.2525
7	0.4	0.03	2.8973	-0.1027

Two friction parameter identification trajectories are generated (as shown in Fig. 2). The first track is used for friction parameter identification and the second track is used to verify the accuracy of friction parameters.

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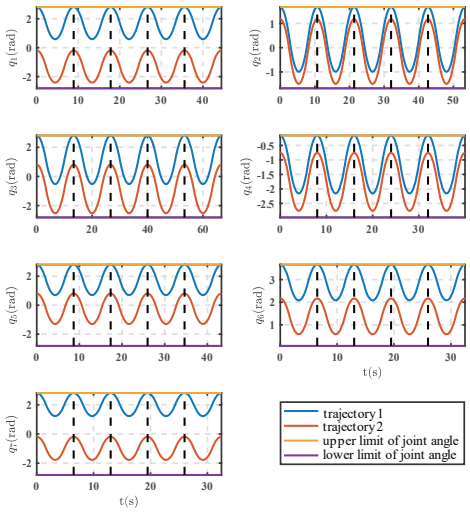


Figure 2 Joint friction parameter identification trajectory

The robot moves according to the designed friction identification trajectory, and a total of 645,800 sets of robot joint angles, angular velocities and link side torque data are collected. Since the viscous friction force of the Panda robot can be ignored [2], the corresponding term in Eq. (13) is discarded. Therefore, the description parameters of the nonlinear friction model (13) can be simplified as $\alpha_{k,i} \in \mathbb{R}, i = 1, 2, \cdots, 5$. The identification of friction parameters can be expressed as the nonlinear least squares problem shown in Eq. (14), which is solved by the Levenberg-Marquardt method. The identified friction parameters are shown in Tab. 2.

TABLE II PANDA ROBOT FRICTION PARAMETERS

joint	$\alpha_{k,1}$	$\alpha_{k,2}$	$\alpha_{k,3}$	$\alpha_{k,4}$	$\alpha_{k,5}$
1	189.582	9.979	-9.933	378.922	-9.951
2	0.851	10.999	4.677	0.408	2.032
3	0.438	11.591	4.351	0.311	2.270
4	21.930	7.363	7.209	0.381	3.401
5	2.958	7.103	5.831	0.467	2.010
6	0.254	2.379	11.894	0.193	2.095
7	179.711	7.843	7.826	0.456	2.881

B. Comparison of Different Friction Models

For the linear friction model, the friction torque of the k joint can be expressed as:

$$\tau_{k,f}(\dot{q}_k) = f_{k,v}\dot{q}_k + f_{k,c}\,\mathrm{sgn}(\dot{q}_k) + f_{k,o}, \tag{23}$$

where $f_{k,v}, f_{k,c}, f_{k,o} \in \mathbb{R}$ are the viscous friction coefficient, Coulomb friction coefficient and Coulomb friction bias coefficient of the k joint, respectively.

For the nonlinear friction model, this paper adopts the sigmoid friction model proposed by Gaz [2], and its friction torque is expressed as:

$$\tau_{f,j}(\dot{q}_j) = \varphi_{1,j} / \left(1 + e^{-\varphi_{2,j}(\dot{q}_j + \varphi_{3,j})}\right), \tag{24}$$

the friction characteristics are determined by $\varphi_{1,j}$, $\varphi_{2,j}$ and $\varphi_{3,j}$.

The three friction models are identified using the data of the first trajectory, and compared using the data of the second trajectory, and the results are shown in Fig. 3. The experimental results show that the nonlinear friction model used in this paper has a better fitting effect than the linear friction model, and is stable at high speed.

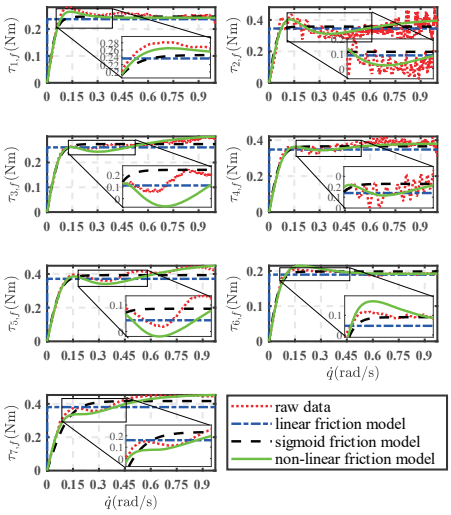


Figure 3 Comparison of different friction models on verification trajectories

When the root-mean-square (RMS) error between the measured and calculated friction moments is used as an evaluation index, the RMS of the nonlinear friction model used in this paper is reduced by 74.06% on average compared to the linear friction model and 32.38% on average compared to the sigmoid friction model, as shown in Figure 4.

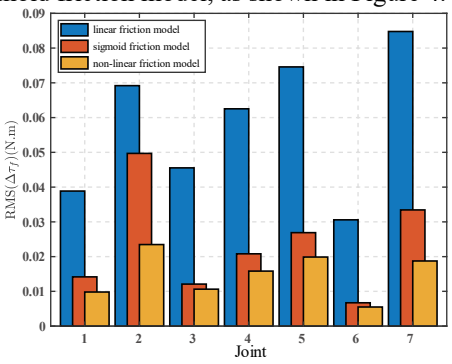


Figure 4 Comparison of RMS errors of different friction models

V. EXPERIMENT 2: IDENTIFICATION AND VERIFICATION OF INERTIAL PARAMETERS

A. Excitation trajectory design

The identification excitation trajectory of the inertial parameter is generated using Eq. (15). Set the angular frequency as $\omega_f = 0.15\pi$, the harmonic order as $L = 5$, and use different $a_{k,l}$, $b_{k,l}$, and $q_{k,0}$, to generate four excitation trajectories that satisfy the constraints of each joint angle, angular velocity, angular acceleration, and workspace, as shown in Fig. 5.

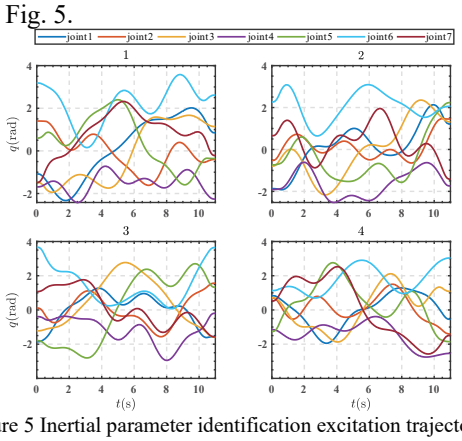


Figure 5 Inertial parameter identification excitation trajectories

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B. *Full parameter identification experiment*

The joint data corresponding to the four excitation trajectories obtained after filtering are divided into two groups: the joint data corresponding to the first trajectory is used for the identification of inertial parameters, and the joint data corresponding to the other three trajectories are used to test the accuracy of the identification results. Substitute the first piece of excitation trajectory data into Eq. (16), and stack all the data to get overdetermined linear equations $\boldsymbol{b} = \boldsymbol{A}\boldsymbol{\phi}$. The nominal value $\boldsymbol{\phi}_0$ of the robot inertial parameters in the constraint regularization weighted least squares formula (19) and the boundary ellipsoid $\boldsymbol{Q}_k$ in the constraint formula (20) are obtained from the CAD model of the Panda robot.

First, the parameter identification problem is solved without regular term and weighting matrix, and the weighted matrix is obtained $\boldsymbol{\Sigma}$. Then, set the regular term coefficient $\gamma = 2$, and solve the complete problem (19) to get the final inertia parameter $\hat{\boldsymbol{\phi}}_*$, as shown in Tab. 3.

At the same time, to quantify the validity of the obtained inertial parameter value $\hat{\boldsymbol{\phi}}_*$, the calculated torque value $\hat{\boldsymbol{\tau}}_*$ and $\hat{\boldsymbol{\tau}}_0$, the RMS of the residuals $\Delta\boldsymbol{\tau}_*$ and $\Delta\boldsymbol{\tau}_0$ of the measured torque $\boldsymbol{\tau}$ are used as indicators, and the data collected from the second, third, and fourth excitation trajectories are used for verification. The comparison data also includes the RMS data of the residual $\Delta\boldsymbol{\tau}$ between the calculated and measured torque in the literature [2], which is also to identify the dynamic parameters of the Panda robot. The result is shown in Fig. 6. The experimental results show that the calculated moment RMS error of the inertial parameters identified in this paper is reduced by 47.08% on average compared with the calculated moment RMS error of the nominal value and by 56.38% compared with the calculated moment RMS error of the reference [2].

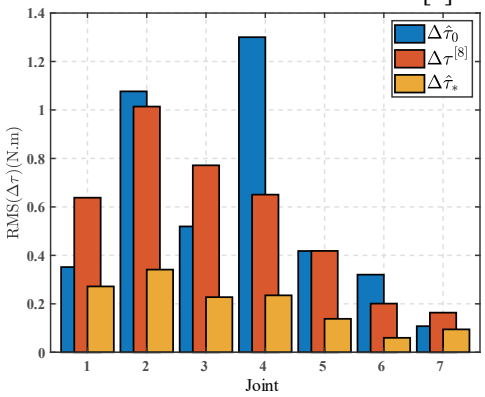


Figure 6 Comparison of RMS error between calculated torque and measured torque of different inertia parameters

TABLE III PANDA ROBOT INERTIA PARAMETERS

Inertia parameter	link1	link2	link3	link4	link5	link6	link7
$m_k\,(\mathrm{kg})$	2.740	2.738	2.335	1.942	2.575	1.966	0.677
$h_{k,x}\,(\mathrm{kg\cdot m})$	9.637E-20	-1.676E-03	1.082E-01	-6.708E-02	8.360E-04	9.903E-02	7.455E-03
$h_{k,y}\,(\mathrm{kg\cdot m})$	-8.901E-02	-1.877E-01	7.451E-02	6.592E-02	1.452E-01	1.042E-02	7.425E-03
$h_{k,z}\,(\mathrm{kg\cdot m})$	-1.852E-01	8.828E-02	-7.464E-02	6.146E-02	-2.914E-01	2.256E-02	4.462E-02
$I_{k,xx}\,(\mathrm{kg\cdot m^2})$	3.345E-02	3.400E-02	1.247E-02	1.217E-02	6.746E-02	3.346E-03	3.910E-03
$I_{k,yy}\,(\mathrm{kg\cdot m^2})$	2.843E-02	9.061E-03	1.729E-02	1.250E-02	5.810E-02	9.289E-03	3.910E-03
$I_{k,zz}\,(\mathrm{kg\cdot m^2})$	9.099E-03	2.902E-02	1.551E-02	1.479E-02	1.224E-02	1.062E-02	8.809E-04
$I_{k,xy}\,(\mathrm{kg\cdot m^2})$	3.026E-21	-1.155E-04	-5.936E-03	5.741E-03	-4.693E-05	-9.604E-04	-2.041E-04
$I_{k,yz}\,(\mathrm{kg\cdot m^2})$	-1.341E-03	1.326E-03	2.066E-04	2.689E-04	9.808E-03	1.051E-05	-5.827E-04
$I_{k,xz}\,(\mathrm{kg\cdot m^2})$	5.414E-21	5.395E-05	1.441E-04	-2.850E-04	8.292E-05	-5.075E-04	-4.514E-04

C. *Physically-consistent Constraint Comparison Experiment*

In order to verify the validity of the full parameter identification method for physically consistent dynamics proposed in this paper, problem (19) is solved without the constraints of physically-consistent. Verify the inertial parameters obtained by identification to determine whether the physically-consistent constraints of Eq. (20) are satisfied, and compare with the verification results of the inertial parameters obtained under the physically-consistent constraints above, as shown in Tab. 4. The experimental results show that the mass of the robot link solved without physically-consistent constraints exceeds the normal range.

TABLE IV PHYSICAL CONSISTENCY COMPARISON OF INERTIAL PARAMETERS

Joint	w/o pc constraints			with pc constraints		
	m_k	$\boldsymbol{I}_k^B \succ 0$	triangle inequality	m_k	$\boldsymbol{I}_k^B \succ 0$	triangle inequality
1	14.505	×	×	2.740	√	√
2	14.761	√	×	2.738	√	√
3	3.489	√	×	2.335	√	√
4	3.493	√	×	1.942	√	√
5	1.668	×	×	2.575	√	√
6	2.290	×	×	1.966	√	√
7	0.085	×	×	0.677	√	√

The RMS error of the calculated moment and the measured moment of the inertial parameters obtained under the condition of no physically-consistent constraints are compared with the inertial parameters obtained above, as shown in Fig. 7. The experimental results show that the inertia parameters satisfying the physically-consistent constraints can make the calculated torque of most joints have the smallest error under the verification trajectory.

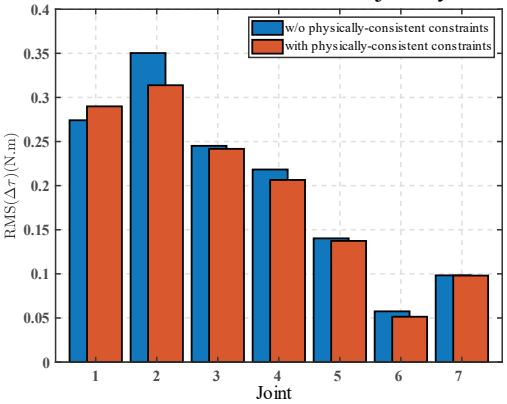


Figure 7 RMS error comparison of solution results with or without physically-consistent constraints

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D. Contrast experiment of simultaneous identification and separate identification

Different from the separate identification proposed in this paper, simultaneous identification of inertial and friction parameters is a common method for identifying robot dynamics parameters. Currently, most simultaneous identification methods need a linear friction model to realize the simultaneous identification of friction and inertia parameters. The simultaneous identification method compared in this paper adopts the linear friction model (24). It ignores the viscous friction force according to the friction characteristics of the Panda robot joints, and the inverse dynamics model of the robot is expressed as:

$$\boldsymbol{\tau} = \boldsymbol{Y}(\boldsymbol{q},\dot{\boldsymbol{q}},\ddot{\boldsymbol{q}})\boldsymbol{\phi} + \boldsymbol{Y}_f(\dot{\boldsymbol{q}})\boldsymbol{\phi}_f, \tag{25}$$

where $\boldsymbol{Y}_f(\dot{\boldsymbol{q}}) = \left[\operatorname{diag}(\operatorname{sgn}(\dot{\boldsymbol{q}}))\right] \in \mathbb{R}^{7\times 7}, \boldsymbol{\phi}_f \in \mathbb{R}^7$.

Then the optimization problem can be expressed as:

$$\min_{\boldsymbol{\phi},\boldsymbol{\phi}_f} \|\boldsymbol{A}\boldsymbol{\phi} + \boldsymbol{A}_f\boldsymbol{\phi}_f - \boldsymbol{b}\|^2_{\Sigma^{-1}} + \gamma\|\boldsymbol{\phi} - \boldsymbol{\phi}_0\|^2_2 \tag{26}$$

$$\text{s.t. } \boldsymbol{P}(\boldsymbol{\phi}_k) = \begin{bmatrix} (\operatorname{tr}(\boldsymbol{I}_k^B)\mathbf{1}_3/2 - \boldsymbol{I}_k^B) & \boldsymbol{h}_k \\ \boldsymbol{h}_k^{\mathrm{T}} & m_k \end{bmatrix} \succ 0,$$

$$\operatorname{tr}(P(\boldsymbol{\phi}_k)Q_k) \geqslant 0, k = 1, \cdots, 7$$

$$\boldsymbol{\phi}_f \geqslant 0 \tag{27}$$

where $\boldsymbol{A}_f = \left[\boldsymbol{Y}_f(\dot{\boldsymbol{q}}_1)^{\mathrm{T}}, \ \cdots, \ \boldsymbol{Y}_f(\dot{\boldsymbol{q}}_M)^{\mathrm{T}}\right]^{\mathrm{T}} \in \mathbb{R}^{7M\times 7}$.

Using the 1st, 2nd, 3rd, and 4th excitation trajectories as sample data, MOSEK and CVX solvers are used to solve the convex optimization problem (27). Use the same method as when solving problem (19) to obtain the weighting matrix $\boldsymbol{\Sigma}$, and finally solve to obtain the inertia parameter $\hat{\boldsymbol{\phi}}_{\#}$.

In order to reflect the friction characteristics at low speed, sinusoidal functions of different amplitudes and periods are used as new verification trajectories, so that the joint velocity crosses zero multiple times, as follows:

$$\dot{\boldsymbol{q}}(t) = \left[A\sin\left(\frac{2\pi}{T}t\right) \ \cdots \ A\sin\left(\frac{2\pi}{T}t\right)\right]^{\mathrm{T}}_{1\times 7}, \tag{28}$$

where A is the amplitude and T is the period. Four trajectories with different speeds are generated, and the parameters are shown in Tab. 5.

TABLE V LOW-SPEED EXPERIMENTS TO VERIFY TRAJECTORY PARAMETERS

Trajectory k	1	2	3	4
A	0.4	0.3	0.2	0.1
T	2	2.5	4	5

The calculation moments $\hat{\boldsymbol{\tau}}_{\#}$ and $\hat{\tilde{\boldsymbol{\tau}}}_{*}$ of the simultaneously identified inertial parameter $\hat{\boldsymbol{\phi}}_{\#}$ and the separately identified inertial parameter $\hat{\boldsymbol{\phi}}_{*}$ are respectively calculated. For each verification trajectory, the evaluation index adopts the sum of the RMS errors of the calculated torque and the measured torque of the seven joints, as shown in Fig. 8. The sum of the RMS errors of the seven joints of the calculated torque $\hat{\tilde{\boldsymbol{\tau}}}_{*}$ on the four different verification trajectories are all smaller than the calculated torque $\hat{\boldsymbol{\tau}}_{\#}$, and the average error is reduced by 19.29%. The experimental results show that the joint friction parameters identified by the nonlinear friction model can reflect the friction characteristics of the joint at low speed and high speed simultaneously, and the error of the calculated torque is smaller.

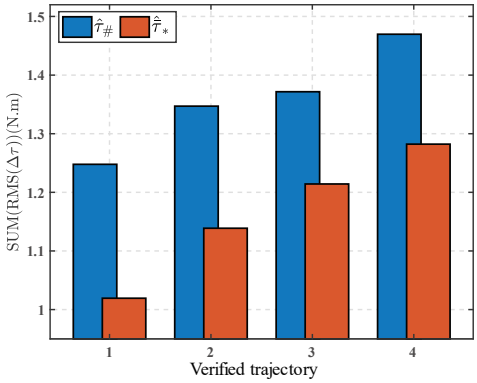


Figure 8 Comparison of the sum of RMS errors of measured torque and calculated torque of each joint under simultaneous identification and separate identification

VI. CONCLUSION

This paper proposes a physically consistent dynamic full-parameter identification method based on a nonlinear friction model. Compared with the linear friction model and the sigmoid nonlinear friction model, the nonlinear friction model used in this method has a minor fitting error, and can better reflect the Stribeck friction effect at low speed; the joint friction parameters obtained through separate identification, realize the decoupling of inertial parameters and friction parameters, and avoids the need to use a linear friction model and the mutual coupling of the two in the simultaneous identification; and the full parameter set obtained by identification can describe the robot dynamics more comprehensively than the minimum inertial set. It can realize more efficient dynamic calculations to be applied to a broader range of scenarios.

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